

CASE STUDY 108 · UNSUPERVISED LEARNING + COSINE
SIMILARITY

PLAYER SIMILARITY ANALYSIS USING MACHINE LEARNING

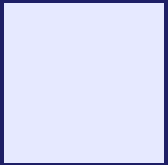
MACHINE LEARNING MAJOR PROJECT



THE SCOUTING PROBLEM

Sports scouting needs a faster way to
compare players

From large-scale statistics to defensible shortlists.



Compare
models
Compare
clustering
algorithms
and select
the
strongest
approach



Recommend
alternatives
Recommend
Top 5
statistically
similar players
for scouting
and
recruitment



Analyze at
scale
Analyze
football
player
statistics at
scale



Discover
groups
Discover
natural
player groups
without
labeled
outcomes

DATA FOUNDATION

One player-level profile replaces fragmented records

2,854

raw records

267

original columns

2,702

unique players after aggregation

39

final performance features

Multiple records belonging to the same player were aggregated into one statistical profile.

01

Clean data

02

Aggregate
by player

03

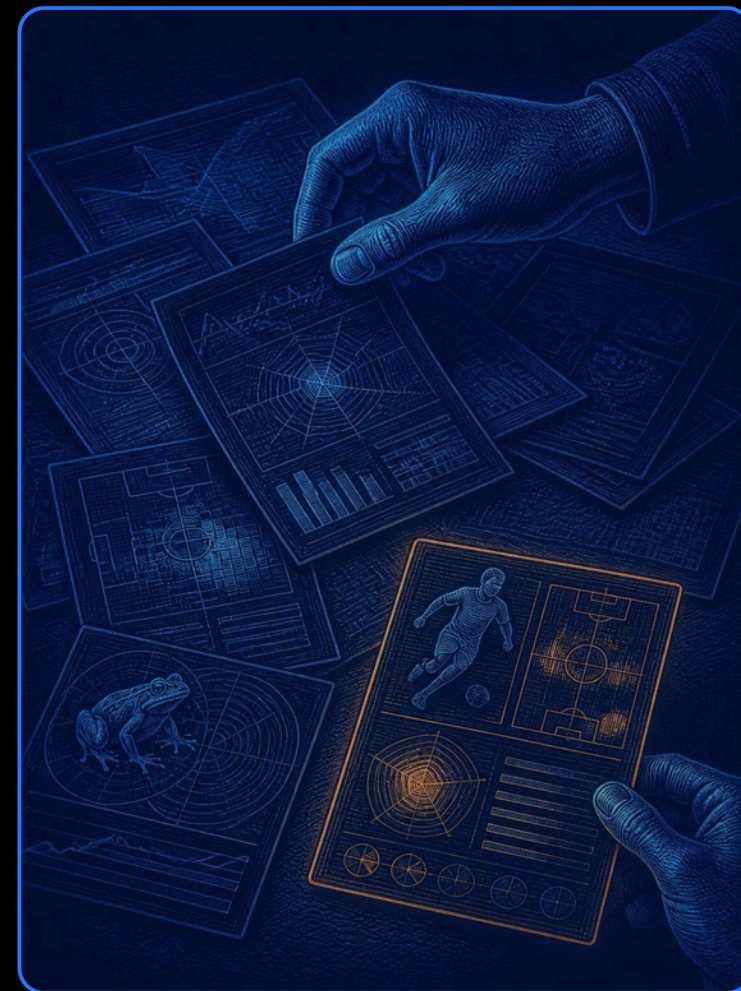
Select
features

04

Handle
missing
values

05

Standardize
features



EXPLORATORY DATA ANALYSIS

The dataset captures varied football profiles

Key takeaway: the dataset spans diverse positions, and performance statistics.



- Player position
- distribution Age
- distribution
- Goals and assists
- distribution Top players by goals
- Correlation heatmap where space allows

METHODOLOGY

The pipeline turns raw statistics into recommendations

1. Football player dataset
2. Cleaning and player-level aggregation
3. 39 performance features
4. Feature scaling
5. PCA for dimensionality reduction and visualization
6. Clustering and model evaluation
7. Cosine similarity
8. Top 5 similar players

Problem type:
Unsupervised
machine learning

PCA VISUALIZATION

PC1 =

49.36%

PC2 =

19.01%

Combined =

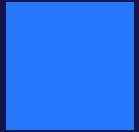
68.38%

MODEL DESIGN

Four models reveal different player-group structures



K-Means: Groups players around cluster centroids



Agglomerative: Builds hierarchical groups of similar players



DBSCAN: Finds dense clusters and identifies noise



Gaussian Mixture: Uses probabilistic distributions to form clusters

PCA: dimensionality reduction and visualization, not a clustering algorithm.

EVALUATION

K-Means leads across all evaluation metrics

Silhouette Score

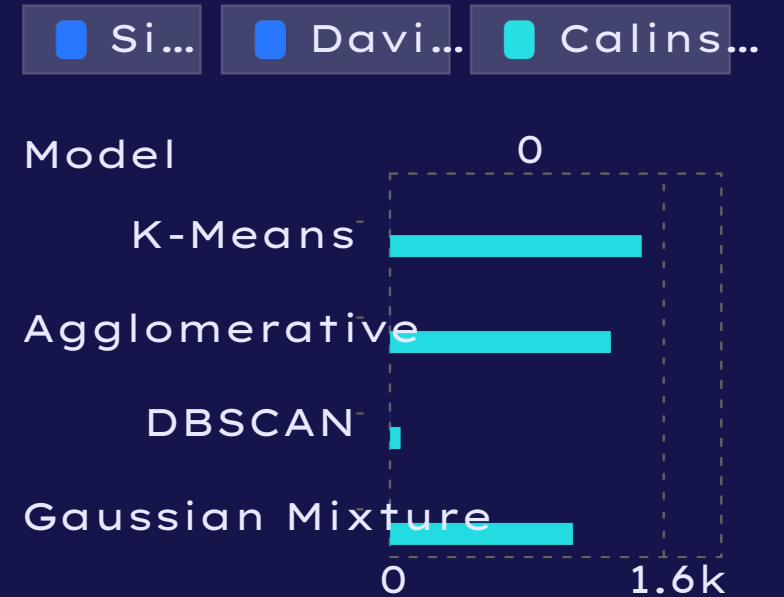
higher is better

Davies-Bouldin Index

lower is better

Calinski-Harabasz Score

higher is better



BEST MODEL

K-Means · K = 2

PLAYER ALTERNATIVES

Cosine similarity finds statistical player alternatives

It compares standardized performance vectors to measure how similar player profiles are.

1. Select a player
2. Use scaled performance features
3. Calculate cosine similarity
4. Rank similarity scores
5. Return Top 5 similar players

A score closer to 1 means a more similar statistical profile. It is not prediction accuracy.

INTERACTIVE DEMONSTRATION

The Streamlit app makes the pipeline interactive



Dashboard



Player Analytics



PCA Visualization



Player Similarity



ML Models and
Model Comparison



Dataset Explorer



01 Select
player

02 View profile
and cluster

03 FindTop5
similar
players

04 Compare
performance

The application is a demonstration layer over the machine learning pipeline, not a claim of online deployment.

CONCLUSION

Machine learning turns football statistics into scouting support

- 2,702 players analyzed across 39 performance features
- Four clustering algorithms compared K-Means selected as the best overall model
- PCA reduced the feature space for visualization
- Cosine similarity identifies Top 5 similar players

FUTURE SCOPE

Position-specific similarity, historical season comparison, contextual features, advanced player embeddings, online Streamlit deployment, and scouting shortlist generation.

THANK YOU